

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building production-grade Node.js and TypeScript applications at scale. Led zero-downtime cloud migrations serving millions of users, architected real-time data pipelines with GCP Pub/Sub, and delivered AI-powered automation solutions. Proven track record of rapid prototyping (0 to 1) and converting to robust, maintainable code. Strong proficiency in SQL databases, modern backend architectures, and AWS/GCP cloud infrastructure. Passionate about applying AI and LLM technologies to solve high-impact problems.

CORE COMPETENCIES

Node.js / TypeScript

Backend Architecture

GCP Cloud Infrastructure

Real-time Data Pipelines

SQL & NoSQL Databases

Rapid Prototyping (0->1)

AI/LLM Integration

AWS Services

WORK EXPERIENCE

MTailor

September 2022 - Present

Software Engineer (US Remote)

- **Led end-to-end migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to CTO
- Built **real-time two-way sync pipeline** between MongoDB and Firestore using Node.js and Google Pub/Sub for reliable message processing
- Developed 3D visualization feature using video overlay and ffmpeg, **increasing buyer conversion**
- Implemented Express Shipping feature, **generating additional \$40 revenue per order**
- **Saved \$5,000/month** by migrating all services from AWS to GCP cloud infrastructure
- Migrated file storage from Amazon S3 to GCS via Python script with maintained access patterns

MTailor

March 2022 - July 2022

Contractor (US Remote)

- Built migration script to read data from MongoDB and write to Firestore with complex processing logic
- Wrote documentation for engineers on working with Firebase SDKs (Node.js and Python)
- Trained Ops team on Firebase Dashboard and backup procedures

Dr Wealth

August 2021 - January 2022

Software Engineer (Singapore Remote)

- Led optimization of PWA delivering **real-time stock market prices** from Firebase backend
- Built ad-hoc jobs to query **2+ million Firestore records** for real-time customer pricing from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js

Mind2matter.co

January 2022 - July 2022

Software Engineer (US Remote)

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines

CodeBits

January 2020 - July 2021

Software Engineer (Uganda Remote)

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- **Architected microservices backend** for FIDA Uganda case management using Apache Kafka, Docker, and Kubernetes
- Developed cross-platform mobile app for LASPNET using React Native and Expo with Firebase push notifications
- Built USSD service for legal aid providers communicating with Node.js backend via gRPC

PROJECTS

Healthcare WhatsApp AI Chatbot AI/LLM

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API. Enabled patients to receive prescription information through conversational AI interface.

Node.js, OpenAI API, Google Vision, WhatsApp Business API

Real-time Pub/Sub Sync Pipeline Data Pipeline

Built live MongoDB to Firestore synchronization pipeline supporting parallel production traffic during cloud migration. Zero data loss with real-time consistency.

Node.js, Google Pub/Sub, MongoDB, Firestore, TypeScript

Marketplace Recommendation Engine Backend

Built collaborative + content-based filtering engine in MongoDB for service provider matching in marketplace platform.

Node.js, MongoDB, Python

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)

2018 - 2022

Kampala, Uganda

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go (learning) **Backend:** Node.js, Express.js, Firebase Cloud Functions, REST APIs, gRPC

Databases: Firestore, MongoDB, PostgreSQL, SQL databases **Cloud:** GCP (Pub/Sub, Cloud Functions, GCS), AWS (S3, EC2, EBS)

DevOps: Docker, Kubernetes, Apache Kafka, NATS Streaming **Frontend:** React, React Native, HTML, CSS, Tailwind

AI/ML: OpenAI API, Google Vision API, LLM Integration